SAKLOLO.: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter highlights the critical challenges currently hindering effective disaster management in Bulacan and CAMANAVA, specifically focusing on the inefficiency of traditional response mechanisms in processing the massive influx of unstructured digital communications during disasters and emergencies. This provides the contextual background and foundation for implementing this computational solution within the local disaster response infrastructure.

1.1 Background of the Study

During severe hydrometeorological events, local disaster management offices are consistently overwhelmed by a massive influx of unorganized, high-velocity digital communications from affected citizens. While these localized text feeds contain critical, actionable information regarding immediate rescue needs and structural damage, the sheer volume and unstructured nature of the data render manual processing highly inefficient. To address this information bottleneck, modern computational semantics relies on vector space mechanics to mathematically transform qualitative narrative posts into numeric matrices. As established by (Mihunov et al., 2022), mapping unstructured disaster text into a high-dimensional geometric space allows automated systems to systematically extract abstract semantic relationships. This mathematical transformation reduces chaotic crisis communications into distinct spatial coordinates, enabling the systematic and automated categorization of emergency surveillance data.

Building upon this mathematical vectorization, recent computational research has suc-

cessfully deployed probabilistic generative models to filter, group, and route these high-volume disaster reports in real time. Previous empirical studies demonstrate that algorithms such as Latent Dirichlet Allocation (LDA) effectively manage the linguistic noise, broken syntax, and data sparsity inherent in online text feeds during acute natural hazards. For instance, during the 2023 Hebei Floods, (Zang, 2026) utilized LDA topic modeling to process unorganized short-text feeds, proving that probabilistic topic distributions can successfully categorize heterogeneous public discourse into actionable, distinct thematic clusters. By treating each text report as a probabilistic mixture of latent topics rather than relying on rigid string-matching, these models allow crisis managers to bypass information overload and automatically isolate urgent, localized operational needs from general baseline noise.

While these computational frameworks have demonstrated profound efficacy in international applications, their deployment is most critical in regions facing compounding, high-frequency environmental crises. The necessity for automated, real-time sensemaking tools becomes acutely apparent when contextualizing the hyper-vulnerable geographic landscape of the Philippine archipelago. In an environment where municipal emergency management offices are repeatedly overwhelmed by simultaneous localized crises, the rapid algorithmic extraction of actionable disaster data is not merely a logistical optimization, but a fundamental requirement for survival and resilience. This operational urgency is directly driven by the country's severe climatic reality.

Disaster-related risks in the Philippines are numerous, as the country is heavily exposed to natural disasters. It is ranked eighth as the most affected globally by extreme weather events, and at least seventy-four (74)% of Filipinos are vulnerable to multiple hazards. Among these hazards, the most catastrophic are tropical cyclones, flooding, storm surges, droughts, earthquakes, and volcanic eruptions. Climate change has caused these events to increase in intensity, as over the past five years the country has experienced destructive typhoons. These disasters have caused economic losses and casualties (World Bank Group, 2023).

Key challenges in mitigating these risks are the lack of coordination and insufficient capacity in implementing Disaster Risk Management (DRM) programs, as most lack integration and cohesiveness to respond to the needs of those exposed to disasters. Major causes are the lack of execution capacity of relevant authorities to efficiently mitigate and address adverse events; the limited coordination between relevant authorities to tackle the increasing magnitude and frequency of disasters; and the overlapping functions, competencies, and duplications of roles causing confusion, delays, and resource misallocations. A core concern is that the timeline of post-disaster response takes a long time and delays rehabilitation and recovery (World Bank Group, 2023).

While social media is utilized for disaster response, the way it is utilized, and in what context, might point to a more nuanced landscape regarding its limitations and the nature of its usage by residents in a certain area. A literature review by (Inoferio et al., 2025) found that social media disaster response is not uniformly developed for all aspects, with man-made disasters not as developed compared to natural disasters in utilization, while COVID-19 and vegetation fires have not been explored as a means for disaster response.

It also found an interesting phenomenon regarding how social media was used for disaster response during disasters. Its utilization was heavy during response and recovery and minimal during preparation. Misinformation, noise, and infrastructure limitations are a concern. It was found to be highly effective in coordinating recovery and providing updates, provided they are filtered between the informative and the controversial (Inoferio et al., 2025)—possibly implying that sophisticated ways to process information and deliver actionable insights to stakeholders could help communities become more resilient and responsive in conducting disaster response.

To bridge this gap, this study proposes *SAKLOLO.: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. This framework introduces a novel hybrid architecture (**System A**) that combines Count Vectorization, Latent Dirichlet Allocation (LDA), and pre-trained XLM-RoBERTa-based

contextual representations into a Joint Matrix. This approach preserves mathematical interpretability while achieving high classification accuracy on chaotic, mixed-syntax Taglish text. Furthermore, the study conducts a rigorous comparative benchmark of this lightweight system against a state-of-the-art open-weight dense transformer (**System B**: `Qwen3-1.7B-Instruct-Q4_K_M`) operating under few-shot prompting conditions, which is also expected to yield a joint matrix.

The study is relevant to modern-day climate risks and high social media penetration in the Philippines, as understanding the digital sentiments and operational needs of vulnerable communities is crucial. This automated processing allows for a tailored response to specific community needs rather than a generalized strategy, enabling informed resource allocation of goods, effort, and time. By benchmarking deterministic and probabilistic machine learning models against generative AI baselines, the study ensures that disaster management offices can focus on their core competencies—dispatching aid and saving lives—rather than manually parsing through an overwhelming influx of digital noise.

1.2 Statement of the Problem

This study evaluates and compares the performance of a hybrid joint matrix pipeline as the baseline (**System A**) against a state-of-the-art large language model joint matrix output (**System B**) in processing unstructured disaster-related text to generate actionable sentiment classifications and latent topic discovery for the Bulacan and CAMANAVA regions. Specifically, it seeks to answer the following questions:

1. Under a relativist comparative evaluation framework, how does **System B** (`Qwen3-1.7B-Instruct-Q`
   perform relative to **System A** (LDA + XLM-RoBERTa Joint Matrix), which serves as the relative reference standard, when both categorize the same corpus of disaster-related textual documents into Negative, Neutral, and Positive sentiments, as measured through:

a. Reference-relative Precision;

b. Reference-relative Recall;

c. Reference-relative Macro-F1 Score;

d. Inter-system agreement with Cohen's Kappa.

2. To what extent do the latent topic structures from the joint matrix of both systems show statistically significant association with sentiment classifications, and which system produces the joint matrix with the tighter topic–sentiment dependency, as measured by:

   a. Chi-Squared;

   b. Cramer's V.

3. Is there a statistically significant operational difference between the outputs of **System A** and **System B** when subjected to a blind comparative evaluation by a purposive stakeholder panel (DRRM operational officers and academic technical experts), as measured through:

   a. Stakeholder Agreement Rate;

   b. McNemar's Test (with continuity correction).

1.3 Significance of the Study

SAKLOLO.: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA is a computational information system for disaster response decision-making. It evaluates how unstructured information in disaster situations can be optimally processed using a Hybrid LDA + XLM-RoBERTa Joint Matrix baseline compared against a `Qwen3-1.7B-Instruct-Q4_K_M` joint matrix pipeline. By

extracting meaningful information—including semantic themes and contextual sentiment—from unstructured emergency feeds, the study establishes a mathematically rigorous framework to understand a disaster and respond to it effectively. This study will benefit several individuals and institutions:

Bulacan and CAMANAVA Disaster Response Offices.

The study is intended to aid disaster response offices in organizing and structuring the unorganized disaster-related information they receive through social media posts. By determining whether a lightweight hybrid architecture or a massive open-weight transformer is more operationally viable for joint latent semantic analysis and its respective sentiment, DRRMO employees are provided with an optimized method to see what disaster-related posts exist and what the sentiments regarding these posts are as received by the concerned communities. This enables personnel to better understand localized field problems and provides them with empirical insight when planning response strategies and disseminating information.

Local Government Units (LGUs).

The study assists LGUs in understand- ing emergency situations occurring in their specific jurisdictions by automatically structuring localized reports. Identifying the most urgent issues through sentiment and latent semantic analysis helps LGUs gain insights into the current state and emotions their respective citizens have during natural disasters. This will aid in assessing the disaster situation locally and assist LGUs in making decisions that are more aligned, and in conducting messaging that takes into consideration how their citizens talk about sets of words that have latent themes, and what their sentiment is around those co-occurring sets of words in social media posts.

Emergency Responders.

The study may be beneficial to emergency responders by providing disaster-related information in a more organized format. As opposed to reviewing social media posts as isolated

and unorganized streams of information, responders can evaluate co-occurring words, their sentiments, and how their respective joint matrices relate to each other. This organized intelligence enables emergency responders to make highly informed, emotionally intelligent, and community-aligned messaging that aids disaster response.

Researchers and Future Developers.

The study acts as a mathematical and computational foundation for those interested in natural language processing within high-stress, low-resource languages (Taglish). Specifically, the novel joint matrix approach combining LDA topic probabilities with XLM-RoBERTa contextual embeddings provides reproducible evidence on hybrid architecture performance versus dense large language models. The integration of Bayesian inference, information theory, and comparative deep learning establishes a robust baseline for future researchers pursuing text optimization in Philippine disaster management.

1.4 Scope and Delimitation

This study focuses on the comparative quantitative evaluation of two computational systems designed to perform sentiment and severity analysis on unstructured disaster-related reports. **System A** is a hybrid pipeline utilizing Count Vectorization, Latent Dirichlet Allocation (LDA), and XLM-RoBERTa contextual representations combined into a Joint Matrix as the baseline system of a same-measurement comparative analysis. **System B** utilizes `Qwen3-1.7B-Instruct-Q4_K_M`, an open-weight large language model, operating under few-shot prompting conditions.

The geographic scope of the study covers the province of Bulacan and the CAMANAVA region (Caloocan, Malabon, Navotas, and Valenzuela). The temporal scope of the corpus spans a ten-year period from 2020 to 2025. Data collection is strictly limited to publicly accessible Facebook posts and comments written in Tagalog, English, and Taglish that meet the established inclusion criteria regarding natural disasters (e.g., typhoons, flooding).

The evaluation framework is strictly delimited to specific mathematical and statistical metrics. Classification performance is measured solely using Precision, Recall, Macro-F1, and Cohen's Kappa. Joint Matrix analysis is to be conducted to analytically test the categorical associations of words to their sentiment using Chi-squared, p-values, and Cramer's V. The operational utility of the systems is evaluated through a purposive blind stakeholder panel assessed using Agreement Rates and McNemar's Test.

The study does not involve real-time integration into actual DRRM command centers, nor does it include the physical dispatch of rescue teams. It performs social media post back-end aggregation with respect to the Data Privacy Act of 2012; all sampled posts and comments are anonymized and protect the identity of the Facebook users. **System A** features a pre-trained XLM-RoBERTa (`cardiffnlp/twitter-XLM-RoBERTa-base-sentiment`) that features approximately one million Tagalog social media posts as part of its training dataset alongside vast numbers of other languages. This study will not employ human-annotated sentiments and will be run as-is, with few-shot prompting on **System B** as an exception.

1.5 Definition of Terms

Agreement Rate.

A statistical metric utilized during the blind stakeholder evaluation to determine the proportion of assessments in which the human expert agrees with the system-generated output.

Cohen's Kappa (κ).

A statistical measure used to assess the inter-system non-probabilistic classification agreement between **System A** and **System B**, factoring in the possibility of agreement occurring by chance.

F1-Score / Macro-F1.

A metric that balances precision and recall by taking their harmonic mean. Macro-F1

calculates the score for each ternary sentiment category (Negative, Neutral, Positive) and assigns equal weight to evaluate overall classification performance.

Qwen3.

An open-weight large language model family developed by Alibaba Group that features dense and mixture-of-experts architectures with built-in hybrid thinking modes.

Joint Matrix.

The novel computational architecture in **System A** that mathematically concatenates lexical representations (Count Vectorization), probabilistic topic distributions (LDA), and contextual word embeddings (XLM-RoBERTa) into a single unified matrix for sentiment classification.

Latent Dirichlet Allocation (LDA).

A Bayesian probabilistic generative model used in **System A** to derive underlying topic-related representations and thematic clusters from the processed disaster documents.

McNemar's Test.

A non-parametric statistical test used during the stakeholder evaluation to analyze paired binary results with continuity correction, assessing if there is a statistically significant marginal difference between the outputs of the two systems.

XLM-RoBERTa (Robustly Optimized BERT Pretraining Approach).

A transformer-based machine learning technique utilized in **System A** to extract deep contextual language representations from the unstructured text corpus that can be used in a multilingual setting.

Taglish.

A Philippine code-switching variety characterized by the intra-sentential alternation of Tagalog/Filipino and English lexical and syntactic elements, prevalent in unstructured social media and emergency text feeds.

Few-Shot Prompting.

An inference paradigm utilized by **System B** (`Qwen3-1.7B-Instruct-Q4_K_M`) in which the model performs sentiment and latent semantic analysis based on natural-language task descriptions and is given some task-specific training examples.

UMass Coherence.

A topic coherence metric that measures the document-level co-occurrence of a topic's top-N words. Values are negative; higher values (closer to zero) indicate more coherent topics. Used in this study to select K, the number of latent topics in Latent Dirichlet Allocation.

CHAPTER II

THEORETICAL FRAMEWORK

This chapter outlines the foundations supporting the development of *SAKLOLO.: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. It is grounded in Vector Space Semantics, the Distributional Hypothesis, and Bayesian Generative Theory, justifying the probabilistic extraction of semantic structures from unstructured disaster text. The literature review discusses Count Vectorization mechanics, contextual language representations (XLM-RoBERTa), generative large language models (LLMs), and Latent Dirichlet Allocation (LDA) applications in crisis environments. A synthesis highlights gaps in processing mixed-language (Taglish) emergency feeds and guides the proposed mathematical joint matrix model.

2.1 Related Theories

The conceptual foundation of this research is grounded in Sensemaking Theory, which asserts that in digital crisis environments, offloading the initial chaotic, high-velocity data to automated systems allows human responders to bypass information overload and focus directly on high-level resource allocation (Pezanowski et al., 2022). To mathematically achieve this automated sensemaking, the computational framework relies on the Distributional Hypothesis in computational semantics. (Balkir et al., 2016) explain that this hypothesis assumes words occurring in similar contexts inherently share similar semantic meanings. This provides the theoretical justification for mapping text into vector spaces. It theorizes that even highly disorganized human crisis communication contains latent geometric patterns, allowing related disaster concepts to naturally cluster together when analyzed

algebraically.

Complementing this geometric assumption is the statistical principle of Exchangeability, rooted in De Finetti's theorem. (Blei et al., 2003) utilized this theory to establish that when modeling text data, the joint probability of a sequence of terms is invariant to their specific grammatical order. This provides the core theoretical justification for the "bag-of-words" approach used in probabilistic topic modeling. Because the theory assumes that documents are the end result of a probabilistic generative process rather than rigid grammatical structures, the proposed system is mathematically validated in ignoring the broken syntax and informal structures typical of emergency posts, focusing purely on the statistical distribution of variables to uncover hidden situational categories.

2.2 Review of Related Literature

2.2.1 Computational Frameworks in International Disaster Literature

The theoretical foundational framework for processing unstructured crisis communications relies on computational linear algebra, vector space modeling, and probabilistic generative models. According to (Mihunov et al., 2022), vector space mechanics represent qualitative narrative posts as numeric constructs in high-dimensional matrices to enable systematic computation during disasters. Within this space, a document–term matrix captures documents across unique vocabulary tokens, where raw text is mathematically transformed using Count Vectorization to systematically extract critical crisis keywords for downstream processing.

Complementing these deterministic vector decompositions, probabilistic frameworks model the underlying semantic structure of text. (Blei et al., 2003) introduced Latent Dirichlet Allocation (LDA) as a Bayesian generative probabilistic model that views each document as a multinomial distribution over latent topics, which are themselves multinomial distributions over vocabulary words governed by Dirichlet priors. Expanding on this

by bridging probabilistic topic models with deep learning, (Zang, 2026) highlights the integration of LDA with contextual transformer architectures (such as BERT). Inferring hidden topic assignment variables alongside deep contextual embeddings from observed social text feeds allows for highly accurate, granular identification and tracking of emergent situational themes as rapid disaster events unfold.

2.2.2 Social Media Utilization in Philippine Disaster Risk Reduction

A literature review on the role of social media in disaster response in the Philippines found that it was heavily employed in response to typhoons and earthquakes, but other disasters such as COVID-19, floods, and vegetation fires are not fully explored. The review indicated that the usage of social media was heavy during response and recovery but minimal during preparation. Because misinformation, noise, and infrastructure limitations remain a concern for its utilization as a reliable disaster response tool, a hybrid computational approach is recommended. Social media was found to be highly effective in coordinating recovery and updates, provided the data is computationally filtered to separate the informative from the controversial (Inoferio et al., 2025).

The Philippines has four thematic areas of Risk Reduction and Management. Disaster Prevention and Mitigation is aimed at strengthening community resilience by minimizing exposure to risks and increasing recovery capacity. Disaster Preparedness involves strengthening community and institutional capacity via collaborative policies and systems. Disaster Response focuses on the reduction of preventable casualties and ensures the basic needs of communities are met and restored rapidly. Finally, Disaster Rehabilitation and Recovery deals with the restoration of livelihoods, economic opportunities, and essential infrastructure that supports the physical and psychological rehabilitation of those affected (Inoferio et al., 2025).

2.3 Review of Related Studies

2.3.1 Global Evaluations of Text Processing in Disaster Management

Empirical implementations across Asian and global disaster contexts demonstrate the practicality of combining probabilistic topic modeling with transformer-based deep learning to process high-velocity unstructured social text feeds. (Zang, 2026) utilized the LDA topic modeling algorithm alongside prompt-BERT models to process unorganized short-text feeds during the 2023 Hebei Floods in China, proving that hybrid architectures effectively categorize heterogeneous social discourse into actionable thematic clusters for crisis managers.

Evaluating short-text social streams during major hydrometeorological events further highlights the necessity of strict statistical metric validation. (Anggara et al., 2025) applied LDA topic modeling to Twitter data during the 2024 West Sumatra flash floods, demonstrating that evaluating topic coherence scores (C_U) across various topic ranks (K) isolates the optimal matrix structure, ensuring distinct and semantically meaningful categories without topic overlap.

In Western disaster management contexts, feature extraction and topic modeling have proven vital for real-time situational awareness. (Mihunov et al., 2022) used LDA to analyze disaster impact feeds during Hurricane Harvey, successfully isolating 9 high-priority target topic vectors out of 24 computed themes to directly map physical infrastructure damage, power grid outages, and localized civilian entrapment locations.

Further demonstrating the efficacy of localized text extraction during severe flooding in Southeast Asia, (Choirul Rahmadan et al., 2020) investigated the Jakarta flash flood domain by applying Latent Dirichlet Allocation (LDA) to high-velocity Twitter streams. Their research successfully clustered unstructured text feeds by localized impact, utilizing the algorithm to map explicit target tokens that detailed specific flooded neighborhoods, severe water depth thresholds, and immediate rescue dispatch needs. Because the hydrology and

social media usage in Jakarta closely mirror the urban–rural challenges found in Philippine municipalities, this study proves that probabilistic generative models can segment baseline social media noise from urgent environmental disaster events to directly support local municipal response.

2.3.2 Local Applications of Topic Modeling and Sentiment Analysis in Crisis Studies

In a local context, Visayan scholars from the University of San Carlos of Cebu and Cebu Technological University utilized Latent Dirichlet Allocation (LDA) and the Hierarchical Dirichlet Process (HDP) to model a qualitative risk assessment for disaster risk reduction response. They attained qualitative data via the disaster risk reduction suggestions of Malasakit, a crowd-sourced disaster initiative across barangays focusing on how to service vulnerable populations (Gorro et al., 2021).

Before conducting the LDA, they performed data cleaning to remove special characters, stop words, and strings under four letters. They then vectorized their entries as a computational prerequisite. For LDA, they measured perplexity scores to determine how stable the machine learning model was in the face of uncertainty, where a lower score indicated better performance. After fitting the models, they discovered the general structural trends: solid waste management, relief and emergency planning, and early warning systems were the most heavily weighted topic sets (Gorro et al., 2021). This study provides a quantitative approach to determining the optimal number of topics based on perplexity scores, which is a valuable methodological addition to local studies.

Another local study utilized LDA in the context of the COVID-19 pandemic in Cagayan Valley, placing a specific emphasis on sentiment analysis. The study utilized Facebook posts from government agencies between March and June 2020, applying standard preprocessing protocols. Using a Java package called Mallet, they fed the data into an LDA model to extract core topics. The generated topic sets went through labeling to identify common themes with the help of the Municipal Disaster Risk Reduction and Management Office

(MDRRMO) of Saguday, Quirino, utilizing six annotated tests judged by multiple public and private sector experts. The study identified five overarching themes capturing public sentiment: COVID-19 testing, relief operations, contact tracing, government programs, and community quarantine (Pagudpud, 2020).

A separate study conducted by Silliman University researchers in Dumaguete City sourced qualitative data from Twitter (X) to conduct topic modeling and document clustering regarding localized sentiments. Utilizing the Twitter4j API, the program gathered 99,942 tweets between 2015 and 2016. After preprocessing the mixed-language texts (Tagalog and English) utilizing string-searching and dictionary-matching algorithms (Aho–Corasick and Damerau–Levenshtein Distance), 22,633 valid tweets remained for training and testing (Montenegro et al., 2018). The researchers utilized Gibbs Sampling to incorporate observed data into the LDA sampling process. Finally, they conducted sentiment analysis via Support Vector Machines (SVM) to classify the texts into positive, negative, or neutral sentiments. While the SVM approach proved functional, it highlights the methodological progression toward modern hybrid architectures and joint matrices that provide deeper contextual understanding than classical SVMs (Montenegro et al., 2018).

2.3.3 Related Undergraduate Theses in Bulacan State University

Bulacan State University has been instrumental in the development of technological applications related to disaster management through various undergraduate studies. An example of this is the study entitled *LokaMatika: Two-Way Location Sharing and Automated Emergency Response Systems with Trilateration and Kinematics* (Manalo et al., 2024). This study designed a mobile application utilizing Dart and Flutter that allows two-way location sharing to help users report and track the locations of vulnerable individuals during emergencies. The researchers utilized ISO/IEC 25010 to evaluate software quality with 243 respondents.

Another related study is *READI: A Modern Way of Saving Lives Using Mobile Appli-*

cation: Real-Time Emergency Alerts and Disaster Information (De Guzman et al., 2019). The researchers developed a mobile application providing real-time emergency alerts and awareness materials to the public, emphasizing the importance of mobile technology in improving communication during emergencies.

While both studies successfully provided a technological basis for disaster management, they focused strictly on the manual emergency response and communication aspects. Neither study attempted to implement natural language processing pipelines, such as contextual Joint Matrices or generative AI Large Language Models (LLMs), to automatically analyze, cluster, and score sentiments on the unstructured text data inherent to disaster communications.

2.4 Synthesis of Literature and Studies, Research Gaps, and Justification

Literature at the international, national, and institutional levels demonstrates the effective use of probabilistic modeling, specifically Latent Dirichlet Allocation (LDA) and contextual transformers, in processing unstructured data into highly interpretable thematic clusters.

Although existing studies have demonstrated significant advancements, critical computational gaps remain. The institutional studies reviewed rely primarily on sensor data, manual user inputs, and GPS hardware, leaving them unequipped to automatically parse high-velocity, multilingual disaster reports. Conversely, international studies successfully integrate Natural Language Processing (NLP) and contextual embeddings, but these methodologies have yet to be rigorously benchmarked for localized Philippine deployment. Local Disaster Risk Reduction and Management Offices (LDRRMOs) currently lack the computational infrastructure to process mixed-syntax Taglish text at scale.

The present study seeks to bridge this gap through a novel comparative methodology. Instead of relying on classical classifiers or manual GPS tracking, the proposed framework

introduces a Joint Matrix architecture integrating LDA topic probabilities with XLM-RoBERTa contextual embeddings to preserve interpretability while mapping urgent distress signals. By benchmarking this hybrid system against a state-of-the-art open-weight dense large language model (Qwen3-1.7B-Instruct) operating under few-shot conditions, this research provides the necessary empirical validation to integrate advanced, automated semantic and sentiment modeling of natural disaster-related social media posts.

CHAPTER III

RESEARCH METHODOLOGY

This chapter details the research methodology employed to evaluate and compare the performance of two disaster-related sentiment and semantic analysis models: **System A**, a hybrid LDA + XLM-RoBERTa Joint Matrix System, and **System B**, `Qwen3-1.7B-Instruct-Q4_K_M`, within the context of the Bulacan and CAMANAVA areas from 2020 to 2025. It establishes a comparative quantitative evaluative research design, detailing the data collection and semi-automated preprocessing of the social media corpus, the architectural frameworks of both systems, and the common measurement metrics used to address the three core research questions. Finally, it outlines the purposive selection of the expert stakeholder panel and the ethical guidelines ensuring participant privacy and data integrity.

3.1 Methods of Research Used

3.1.1 Overall Research Design

This study utilizes a comparative quantitative evaluative research design to evaluate the performance of two disaster-related sentiment and semantic analysis systems: **System A**, a hybrid LDA + XLM-RoBERTa Joint Matrix System, and **System B**, `Qwen3-1.7B-Instruct-Q4_K_M`. Both systems will have the same corpus of disaster-related textual data from the Bulacan and CAMANAVA area within the study period. The two systems are designed to be evaluated on multiple complementary measures rather than a single measure of performance. These measures correspond to the three main research questions: classification performance, latent topic–sentiment association analysis, and sampled comparative evaluation of their outputs.

This study follows the common comparative pipeline that collects disaster-related textual data, filters, preprocesses, and prepares it for analysis by both systems. **System A** processes this corpus via Count Vectorization, Latent Dirichlet Allocation (LDA), and XLM-RoBERTa-based contextual application, which are then combined into a joint matrix for further evaluation. **System B** processes the same corpus using `Qwen3-1.7B-Instruct-Q4_K_M` through structured prompts for the same sentiment analysis and semantic assessment. The two systems thus fall under a common data source, allowing for a comparative evaluation to proceed.

The study's textual disaster-related data is geographically linked to Bulacan and CAMANAVA, an area consisting of Caloocan, Malabon, Navotas, and Valenzuela during the study period of 2020-2025. The corpus includes social media content, government or disaster-management reports, relevant news reports, and other accessible public textual sources on Facebook relative to the inclusion and exclusion criteria. The methodology identifies Bulacan and CAMANAVA as the geographic scope and specifies Tagalog, English, and Taglish as the principal language context of the textual data.

The study will apply a common measurement approach to the eligible corpus. Each document is processed by both systems under their respective procedures. This allows the resulting sentiments of both to be compared directly.

3.1.2 Research Question 1: Relativist Comparative Classification Performance

The first question of the study evaluates the sentiment classification performance of **System A** and **System B** under a relativist comparative evaluation framework. In this framework, no absolute ground truth is posited. Instead, **System A**—whose system components (XLM-RoBERTa) were pretrained on human-annotated sentiment data—is designated as the relative reference, and **System B** is evaluated against it on the same corpus of disaster-related documents.

Each document is assigned one of the three sentiment categories: Negative, Neutral, or

Positive. Four metrics are computed:

- **Reference-relative Precision, Recall, and Macro-F1.** These are asymmetric metrics computed for **System B** against **System A** as the relative reference. They measure the extent to which **System B** reproduces **System A**'s classifications.
- **Inter-system agreement with Cohen's Kappa.** This is a symmetric, chance-corrected measure of agreement between the two systems. Unlike the reference-relative metrics, it does not privilege either system as the reference.

Both systems are applied to the same prepared corpus and yield a 3×3 matrix in which rows represent **System A** classifications and columns represent **System B** classifications. Diagonal cells indicate agreement; off-diagonal cells are disagreements. The confusion matrix is the basis of all reference-relative performance metrics.

The relativist framing is intentional, as the study does not claim that **System A**'s outputs are objectively correct. It asks whether a state-of-the-art open-weight LLM under few-shot prompting conditions reproduces what a lightweight hybrid pipeline—treated as the relative reference—produces.

3.1.3 Research Question 2: Joint Matrix Latent Topic Structure and Sentiment Classification Relationship Analysis

The second question of the study evaluates whether the latent topic structures and sentiment classifications from the joint matrix of both **System A** and **System B** demonstrate a measurable association with one another, and how this association describes the dynamics between latent topics and sentiment classifications and the strength of their relationships analytically. The study implements a chi-square significance test to measure whether the relationship between the latent topics and sentiment classifications exists, and a relationship strength test with Cramer's V to measure the true strength of association between latent topics and sentiment classifications and how big and impactful this is.

The independent relationship analysis of both systems is to be conducted with the data inputs and n specifications of potential latent topic themes to create an equally sized joint matrix or contingency table. The analysis shall report relevant relationship measurements of performance: the Chi-Square Statistic and Cramer's V. The comparison emphasis is on determining analytically the existence of a statistical relationship in their respective joint matrices and the strength of their association independently produced by both systems. Then they will be independently assessed against their acceptability thresholds to conclude whether such a relationship exists and how impactful it is on their evaluation of which words are being talked about frequently together and how people feel about them, before comparing the two systems again for further insights. Thresholds apply exclusively to independent analysis.

3.1.4 Research Question 3: Sampled Comparative Evaluation

The third question of the study examines both **System A** and **System B** through a common evaluation corpus. The objective is to attain a representative output of the two systems. It is to be structured to ensure representation of outcome patterns where the two systems agree, where they disagree, and cases representing different sentiment levels, where deemed appropriate.

The sampled documents from both systems are to be blindly evaluated by stakeholders. This assessment will focus on their own assessment of the same documents and their agreement or disagreement when presented in two blind tests sampled from evaluated outputs that do not specify which is made by **System A** and which is made by **System B**. It will be a check-or-cross test. The two systems will be evaluated individually in terms of stakeholder agreement rate and McNemar's test. They will be evaluated independently and together when appropriate for further insights. Acceptability thresholds are to be applied appropriately to non-comparative measures.

3.1.5 Evaluation Framework

The study utilises a same-measurement comparative evaluation framework in which **System A** and **System B** are applied to the same corpus of disaster-related textual data under comparable evaluation conditions. This framework is designed to ensure that differences observed between the two systems are attributable to their respective processing and analytical approaches rather than to the data themselves.

The comparative process begins with the collection and preparation of an eligible disaster-related textual corpus from Bulacan and CAMANAVA. The same processed corpus is then provided to both systems according to their respective procedures. **System A** applies LDA and an XLM-RoBERTa-based joint matrix, while **System B** processes the same textual input via `Qwen3-1.7B-Instruct-Q4_K_M` using a predefined prompting procedure. The resulting sentiment and latent topic outputs are recorded via a common measurement structure established for the study.

The output of both systems is evaluated through three complementary analyses. First, their sentiment classifications are assessed using performance measures associated with Research Question 1. Second, their joint matrices are assessed using performance measures associated with Research Question 2. Third, a randomly selected sample of common outputs is subjected to blind stakeholder evaluation in which their identities are concealed, and their outputs are assessed under the same evaluation conditions.

This framework therefore is a common-input and comparative-analysis sequence. Each system is evaluated either together or independently relative to mathematical viability and consistency of measures of performance to gain insights and findings. This allows the study to evaluate the independent ability of each system and the relative performance of one over the other with respect to the mathematical assumptions of each evaluation framework.

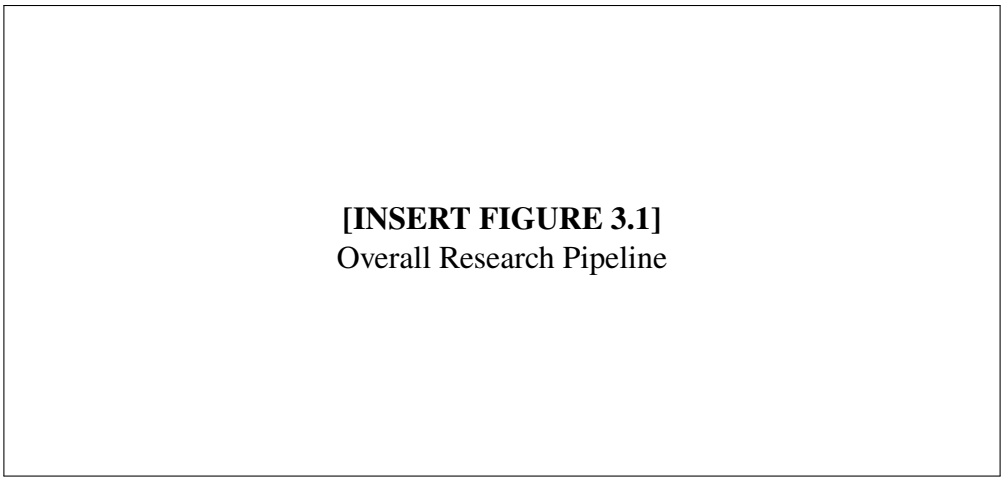


Figure 3.1: Overall Research Pipeline

3.2 Data Sources and Corpus

The textual data will be collected primarily from publicly accessible Facebook posts and comments related to disasters occurring in Bulacan and CAMANAVA during the study period of 2020-2025. Facebook is used as the principal data source because posts and comments provide naturally occurring public expressions that may contain information on public sentiment and perceived disaster severity. The corpus will include textual content written in Tagalog, English, and Taglish.

Data collection will be a semi-automated manual extraction procedure supported by a web scraper browser extension to aid in extracting the textual data from posts and their comment threads. The relevant Facebook posts and their available comments will be identified using the study's own inclusion and exclusion criteria. The web scraper extension will assist in the extraction and organization of available data, while manual procedures will identify relevant sources.

The data sourcing will target an approximate corpus size of 10,000-15,000 records and above. It will source from publicly available pages of disaster-related agencies around Bulacan and CAMANAVA, with optional inclusions of newer agencies and wider public government pages. The timeframe of 2020-2025 was chosen to obtain relevant qualitative

data that is relatively not out of date and is feasible to attain.

3.2.1 Inclusion Criteria

Facebook posts or comments are to be included if they satisfy the following criteria: they are publicly accessible; they are related to a disaster, disaster event, disaster impact, response, recovery, or other related themes; they are relevant to Bulacan or CAMANAVA; they fall within the 2020–2025 study period; they contain textual content suitable for sentiment and semantic analysis; and they are written in Tagalog, English, or Taglish.

3.2.2 Exclusion Criteria

Facebook posts or comments are to be excluded when they are not publicly accessible; when they are not related to a disaster, disaster event, disaster impact, response, recovery, or other related themes; when they fall outside the scope of Bulacan or CAMANAVA; when they fall outside the 2020–2025 study period; when they contain no textual content suitable for sentiment and semantic analysis; when they consist entirely of spam, advertisements, promotional materials, and other unrelated content; when they are duplicated records bearing the same content or corrupted text; and when they contain any mentions, hashtags, private information, or emojis.

Together, the resulting eligible records constitute the common corpus used by both **System A** and **System B** under their individual procedures, so that differences in sentiment and latent topic outputs can be examined under a common data basis.

3.3 Data Preprocessing

The collected common corpus is to undergo preprocessing before being submitted to **System A** and **System B**, with the intention of removing irrelevant content while preserving linguistic information necessary for sentiment and latent topic analysis. The prepared corpus is to be sent to both systems to remain consistent with the comparative analysis framework. The

records are to be screened and cleaned based on the inclusion criteria; any content that falls within the exclusion criteria is to be excluded and removed. Text normalization is to be done next to improve consistency; this will remove formatting artifacts, excessive whitespace, and other characters that are not relevant to the study. Record consolidation is to be done afterward to remove remaining duplicates. Finally, the corpus is to be subjected to textual completeness and analytical suitability checks before being fed to the two systems.

3.4 Comparative Systems

This study evaluates two computational systems for disaster-related sentiment and latent topic analysis: **System A**, a hybrid LDA–XLM-RoBERTa Joint Matrix System, and **System B**, `Qwen3-1.7B-Instruct-Q4_K_M`. Though the two different systems employ different processing approaches, both receive the same prepared disaster-related textual corpus and produce sentiment and latent topic outputs that are evaluated using the common measurement framework of the study. This arrangement allows the study to distinguish differences arising from the analytical approaches of the systems while maintaining a common basis for comparison.

3.4.1 System A: LDA-XLM-RoBERTa Joint Matrix System

System A employs a hybrid approach that combines Count Vectorization, Latent Dirichlet Allocation (LDA), and XLM-RoBERTa-based contextual representations into a joint matrix. Count Vectorization provides the lexical representation of the textual corpus, while LDA is used to derive topic-related representations from the processed documents. XLM-RoBERTa provides contextual language representation for sentiment analysis of the disaster-related text. These representations are integrated into a joint matrix that serves as the computational representation for the subsequent analysis.

Count Vectorization

Count Vectorization is the process of converting text documents into numerical vectors that allow machine learning models to perform natural language processing (NLP). This starts with removing unnecessary elements such as punctuation, URLs, and numbers; all text is also converted to lowercase. The model processes them by splitting the documents into words so that the model can understand and analyze each word individually, followed by identifying unique words from the preprocessed text. This serves as the reference that maps text into a numerical vector. The text is then converted into numerical vectors for the machine learning model to process. It acts as a bridge between human language and machine language (Kozhevnikov & Pankratova, 2020).

Vectorization is a function that maps a text document into a numerical vector in a high-dimensional space. Let

$$\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$$

be a set of N documents, and let

$$V = \{w_1, w_2, \ldots, w_{|V|}\}$$

be the vocabulary of $|V|$ unique words. Then vectorization is a function

$$\phi : \mathcal{D} \longrightarrow \mathbb{R}^{|V|} \tag{3.1}$$

such that each document $d_i \in \mathcal{D}$ is mapped to a vector

$$\phi(d_i) = \left[x_{i1},\ x_{i2},\ \ldots,\ x_{i|V|}\right] \tag{3.2}$$

where x_{ij} represents the weight or value of word w_j in document d_i. For count vectorization,

the value x_{ij} is simply the raw frequency of word w_j in document d_i:

$$x_{ij} = \text{count}(w_j, d_i) \tag{3.3}$$

For example, given the vocabulary $V = \{\text{“Word 1”}, \text{“Word 2”}, \text{“Word 3”}, \text{“Word 4”}\}$:

Table 3.1: Illustrative Count Vectorization Matrix

Document	**Word 1**	**Word 2**	**Word 3**	**Word 4**
d_1	1	1	1	1
d_2	1	1	1	0

such that $\phi(d_1) = [1, 1, 1, 1]$ and $\phi(d_2) = [1, 1, 1, 0]$.

Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation is a generative probabilistic model of a corpus. The assumption is that each document is represented as a random mixture over latent topics, and each of these topics is characterized by a distribution of words. Its generative process follows that, for each document d in the corpus:

1. Choose $\theta_d \sim \text{Dirichlet}(\alpha)$.
2. Choose $\phi_k \sim \text{Dirichlet}(\beta)$ for each topic $k \in \{1, \ldots, K\}$.
3. For each of the N_d words $w_{d,n}$:
    a. Choose a topic $z_{d,n} \sim \text{Multinomial}(\theta_d)$.
    b. Choose a word $w_{d,n} \sim p(w_{d,n} \mid z_{d,n}, \beta)$, a multinomial probability conditioned on the topic.

Each document draws its overall topic mixture θ_d from a Dirichlet distribution parameterized by α:

$$\theta_d \sim \text{Dirichlet}(\alpha) \tag{3.4}$$

Given the parameters α and β, the joint distribution of a topic mixture θ_d, a set of K topics ϕ, and a set of N_d words $\mathbf{w}_d$ is given as

$$p(\mathbf{w}_d, \mathbf{z}_d, \theta_d \mid \alpha, \beta) = p(\theta_d \mid \alpha) \prod_{n=1}^{N_d} p(z_{d,n} \mid \theta_d)\, p(w_{d,n} \mid z_{d,n}, \beta) \tag{3.5}$$

where $z_{d,n}$ is simply one of the unique topics such that $z_{d,n} \in \{1, \ldots, K\}$. Integrating over θ_d and summing over $\mathbf{z}_d$, we obtain the marginal distribution of a document:

$$p(\mathbf{w}_d \mid \alpha, \beta) = \int p(\theta_d \mid \alpha) \left[\prod_{n=1}^{N_d} \sum_{z_{d,n}=1}^{K} p(z_{d,n} \mid \theta_d)\, p(w_{d,n} \mid z_{d,n}, \beta) \right] d\theta_d \tag{3.6}$$

Finally, taking the product of the marginal probabilities of single documents, we obtain the probability of a corpus:

$$p(\mathcal{D} \mid \alpha, \beta) = \prod_{d=1}^{M} p(\mathbf{w}_d \mid \alpha, \beta) \tag{3.7}$$

The parameters of the LDA get modified depending on their level. Overall, the model has three levels: corpus level, document level, and word level. The parameters α and β are corpus-level parameters and are sampled once in the process of generating a corpus. The variables θ_d are document-level variables and are sampled once per document. Finally, the variables $z_{d,n}$ and $w_{d,n}$ are word-level variables and are sampled once per word in each document (Blei et al., 2003).

Latent Dirichlet Allocation (LDA) K-Topic Selection

The number of K latent topics was selected empirically through a coherence sweep across $K \in \{2, 3, \ldots, K_{\max}\}$. For each value of K, the model was fitted on the prepared corpus with a fixed random state of 42, batch learning, and 100 maximum iterations, and the mean UMass Coherence score and its respective criterion determined the K-topic quantity of the model.

The UMass coherence score evaluates the quality of the topic models. It relies on the internal corpus of the model, and words are calculated in order of their frequency within the topic; the expected outcome is negative, with values nearest to zero being better. To compute the UMass coherence score:

$$C_{\text{UMass}}(T_k) = \sum_{m=2}^{N} \sum_{n=1}^{m-1} \log \frac{D(w_m, w_n) + 1}{D(w_n)} \tag{3.8}$$

where $T_k = \{w_1, w_2, \ldots, w_N\}$ is the list of N probable words in topic k, $D(w_m, w_n)$ is the number of documents in the corpus containing both w_m and w_n, and $D(w_n)$ is the number of documents containing w_n. This metric relies on co-occurrence statistics gathered from the corpus being modeled and is domain specific (Mimno et al., 2011).

Because the sentiment category has exactly three categories, the degrees of freedom contribution of Cramer's V is fixed at 2 as long as $K \geq 3$. This ensures that Cramer's V values are comparable across different values of K and across both systems. The selected K—the UMass coherence score nearest to zero among the sampled values—will be the selected amount of K-topics for the LDA configuration and the amount of topics for **System B**'s Stage 1, to preserve the matched dimensionality of their joint matrices required for the comparative evaluation framework.

XLM-RoBERTa (XLM-T)

LDA only gives us a platform to conduct latent semantic analysis; it only allows us to see which words co-occur relative to a given corpus. It does not tell us whether people talk about these sets of words or themes negatively, positively, or neutrally. Sentiment analysis utilizing language models in multilingual settings makes this possible. XLM-T is a transformer-based model that is an extension of XLM-RoBERTa trained on social media posts on X (formerly Twitter). The model is trained on 198 million social media posts that include English, Tagalog, and many more, with Tagalog/Filipino comprising roughly one

million posts (Barbieri et al., 2021).

Following pretraining, XLM-T was fine-tuned for sentiment classification on human-annotated datasets in eight core languages: Arabic, English, French, German, Hindi, Italian, Portuguese, and Spanish. Tagalog/Filipino is not among these eight fine-tuning languages. Instead, Tagalog sentiment classification is supported via zero-shot cross-lingual transfer. Because the pretraining stage learns a shared multilingual representation, sentiment supervision from the eight core languages generalizes to Tagalog without any Tagalog sentiment labels being used during training. This modification allows XLM-T to outperform other multilingual NLP transformer-based models that rely on more formal training data such as Wikipedia and CommonCrawl (Barbieri et al., 2021).

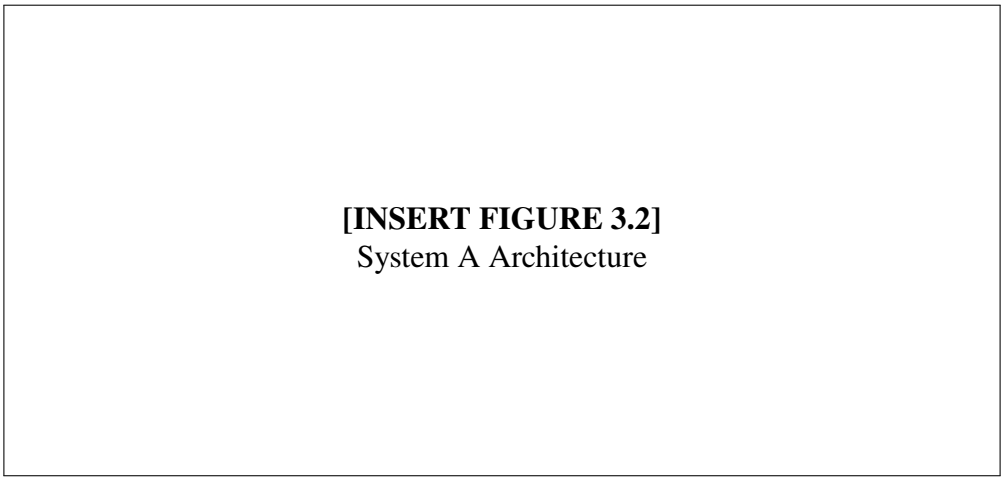


Figure 3.2: System A Architecture

The system generates sentiment classifications using the defined ternary sentiment categories of Negative, Neutral, and Positive, as well as the latent topics and their respective words. These outputs are fixed before the comparative evaluation.

3.4.2 System B: `Qwen3-1.7B-Instruct-Q4_K_M`

System B employs `Qwen3-1.7B-Instruct-Q4_K_M` as a few-shot language model for disaster-related sentiment and latent topic analysis. Unlike **System A**, which is a hybrid approach utilising multiple representations, **System B** processes the textual inputs directly

through a predefined prompting procedure.

The prompting procedure specifies the required sentiment categories and latent topics output format to ensure consistency and that comparative analysis assumptions are respected. For sentiment analysis, the system produces one of the three defined categories: Negative, Neutral, or Positive.

Model Specifications

System B utilizes the Qwen3-1.7B-Instruct model in GGUF format at Q4_K_M quantization. Inference is performed with temperature at 0.1 and a top-p of 1.0. It has a maximum generation length of 48 tokens. This low temperature reduces sampling variance across documents and supports reproducibility of the classification outputs.

Two-Stage Prompting Architecture

System B operates in two stages: topic discovery and joint classification. This separation is crucial, as the model must first establish the canonical topic taxonomy before it can assign topic labels to individual documents. The number is fixed to the value of the K-topic to match **System A**'s LDA dimensionality, which guarantees that the joint matrix (topic–sentiment contingency table) of both systems has the same matrix shape and is legible for the common measurement framework.

Stage 1: Topic Discovery. In the first stage, a representative sample of 200 documents is drawn from the prepared corpus. This representative sample is presented to the model with instructions to synthesize a canonical taxonomy of distinct K topics. The prompt directs the model to focus on core actions, problems, or themes rather than proper nouns, locations, or geographic identifiers. Each topic must have a short label and a set of keywords extracted from the representative sample.

The model returns a JSON array of topic objects. The prompt enforces strict JSON

Listing 3.1: Stage 1 Prompt

```
TOPIC_DISCOVERY_SYSTEM = """
You are an expert qualitative data analyst. Your task is to
    analyze a batch of text documents and synthesize a
   canonical taxonomy of exactly {n_topics} distinct topics
   .

GUIDELINES:
- Focus on the core ACTION, PROBLEM, or THEME (e.g.,
   medical relief, structural damage, supply distribution).
- Strictly omit all proper nouns, locations, regions, and
   geographic identifiers.
- Each topic must have a short 'label' (2-4 lowercase words
   ), and actual 'keywords' (bag-of-words) extracted
   directly from the texts.
- Respond ONLY with a valid JSON array of objects. No
   markdown, no explanations.
- CRITICAL: Your output MUST start with "[" and end with "]
   ".

EXAMPLE OUTPUT:
[
{{"label": "rescue efforts", "keywords": ["rescue", "
   stranded", "volunteer", "bagal", "rescuers"]}},
{{"label": "flood impact", "keywords": ["baha", "tubig", "
   lubog", "taas", "stranded"]}},
{{"label": "health risks", "keywords": ["leptospirosis", "
   sintomas", "check-up", "baho"]}}
]
/no_think
"""

TOPIC_DISCOVERY_USER = """
Here is a representative sample of {n_docs} documents from
   the dataset:

{documents_block}

Generate the JSON array of exactly {n_topics} topics based
   on the themes present in these documents.
JSON Output:
"""
```

output with no markdown or explanations. This discovered taxonomy is validated for structural completeness before being used in Stage 2. The final taxonomy is recorded to a CSV file and is used for the common measurement framework.

The discovery sample is drawn using random sampling from the whole prepared corpus. This rule is fixed to prevent bias toward the first entries in the corpus being sampled first.

Stage 2: Joint Sentiment and Topic Classification. In the second stage, each document in the prepared corpus is classified individually. The prompt presents the model with the final taxonomy from Stage 1 and a small set of few-shot examples demonstrating the sentiment decision rule. The model is required to output a single JSON object with two fields: `sentiment` and `topic`.

The sentiment must be exactly one of the following: `POS`, `NEG`, or `NEU`. The prompt defines the following as:

- `POS`: the text expresses joy, gratitude, hope, relief, or praise;
- `NEG`: the text expresses anger, sadness, fear, frustration, or criticism;
- `NEU`: the text is purely informational, factual, asking for help, or expresses a balanced or mixed view.

The prompt explicitly instructs the model not to assume that factual requests are negative, addressing a common failure mode in disaster text classification. The `topic` field must be exactly one of the label values from the discovered taxonomy. As in Stage 1, the model is instructed to output only valid JSON with no markdown or explanations.

Both systems are therefore applied to the same processed corpus but retain their individual internal procedures. Their outputs are to be subjected to the same evaluation framework for independent assessment and comparative analysis.

Listing 3.2: Stage 2 Prompt

```
JOINT_SYSTEM = """
You are an expert analyst classifying social media posts.
   You must output TWO labels: SENTIMENT and TOPIC.

SENTIMENT must be exactly one of: "POS", "NEG", "NEU".
- POS: The text expresses joy, gratitude, hope, relief, or
   praise.
- NEG: The text expresses anger, sadness, fear, frustration
   , or criticism.
- NEU: The text is purely informational, factual, or
   expresses a balanced/mixed view. Do not assume factual
   requests are negative.

TOPIC must be exactly one of the 'label' values from the
   provided taxonomy list.

Respond ONLY with a valid JSON object. No markdown, no
   explanations.
FORMAT:
{{"sentiment": "LABEL", "topic": "exact topic label from
   list"}}
/no_think
"""

JOINT_USER = """Available taxonomy (label and description):
    {topic_list_with_descriptions}

Examples:
Text: "Salamat sa mga rescuers! Ligtas na kami."
JSON: {{"sentiment": "POS", "topic": "rescue efforts"}}

Text: "Base sa PAGASA, tuloy-tuloy ang ulan hanggang
   weekend. Maghanda po."
JSON: {{"sentiment": "NEU", "topic": "flood warning"}}

Text: "Nakakagalit! Ang bagal ng rescue. Ilang oras na
   kaming stranded."
JSON: {{"sentiment": "NEG", "topic": "flood impact"}}

Text to classify:"{text}"

JSON Response:
"""
```

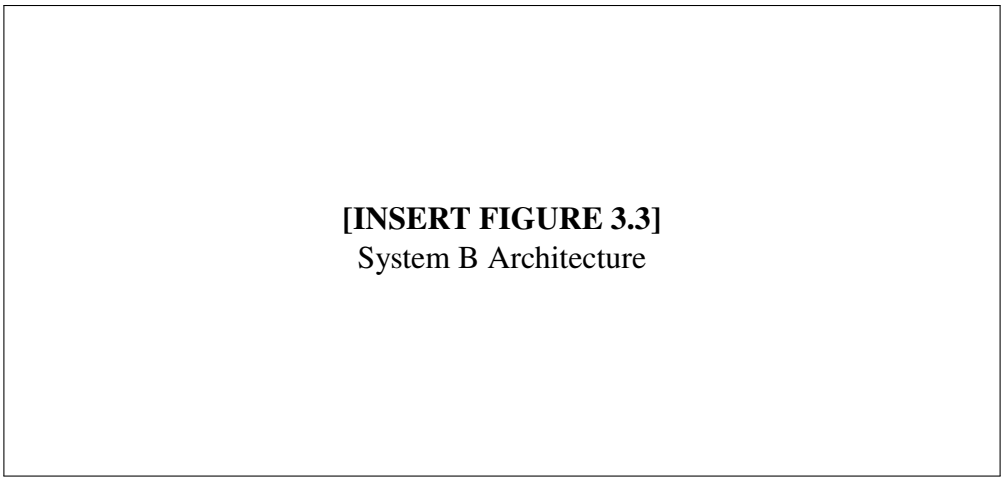


Figure 3.3: System B Architecture

3.5 End Products of System A and System B: The Joint Matrix

In order for both systems to undergo the same measurement comparative analysis, both their latent topic matrices and their sentiment classification matrices must be fused together onto a single joint matrix that has the same dimension and size. This will allow quantifiable measurements of performance to be performed on both systems' outputs independently and relative to one another.

Contingency tables allow the evaluation of the joint distributions, frequency counts, and percentages of two categorical variables. Formally, the table variables define the rows and columns, one variable defining the rows of the table and another variable defining the columns of the table (NCSS, LLC, n.d.).

Let $T = \{\tau_1, \ldots, \tau_M\}$ be the row variable of latent topics with M topics, and $S = \{\sigma_1, \ldots, \sigma_N\}$ be the column variable of sentiment classifications with N classes. Then an

$M \times N$ contingency table with marginal totals is defined as

$$J = \begin{pmatrix} j_{11} & j_{12} & \cdots & j_{1N} & R_1 \\ j_{21} & j_{22} & \cdots & j_{2N} & R_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ j_{M1} & j_{M2} & \cdots & j_{MN} & R_M \\ C_1 & C_2 & \cdots & C_N & n \end{pmatrix} \tag{3.9}$$

where each cell entry is defined as

$$j_{ij} = \left| \{ \, d \in \mathcal{D} : t(d) = \tau_i \ \wedge \ s(d) = \sigma_j \, \} \right| \tag{3.10}$$

with $t(d)$ representing the assigned latent topic of document d and $s(d)$ representing its assigned sentiment classification.

3.6 Measurement and Evaluation

The information collected from the two systems will be processed, coded, and analysed with suitable performance measures that correspond to the three research questions of this investigation. **System A**, the Hybrid LDA + XLM-RoBERTa Joint Matrix System, and **System B**, `Qwen3-1.7B-Instruct-Q4_K_M`, will be assessed using the exact evaluation framework to ensure comparative integrity in the evaluation results. Each statistic will be run separately on each system prior to comparing and making relative findings.

3.6.1 Research Question 1: Reference-Relative Classification Performance

The purpose of Research Question 1 is to assess the overall sentiment classification performance of both **System A** and **System B** as they classify the same set of disaster-related documents. It answers the question of whether a large language model with more parameters and compute can brute-force its way into replicating an established, specialised tool present

in **System A** for both latent semantic analysis and sentiment analysis. Both **System A** and **System B** will assign a sentiment classification to each document in one of three categories: Negative, Neutral, and Positive. The classification accuracy of **System A** and **System B** will be measured in terms of Reference-relative Precision, Recall, F1-Score, Macro-F1, and Cohen's Kappa.

A confusion matrix is to be constructed so that the two systems can be assessed for Research Question 1. Both systems share the same sentiment categories— Negative, Neutral, and Positive—yielding a 3×3 classification matrix:

Table 3.2: Reference-Relative Confusion Matrix (**System A** as Reference, **System B** as Comparison)

System A \ **System B**	**Negative**	**Neutral**	**Positive**
Negative	a_{11}	a_{12}	a_{13}
Neutral	a_{21}	a_{22}	a_{23}
Positive	a_{31}	a_{32}	a_{33}

where rows represent **System A**, columns represent **System B**, diagonal cells represent identical classifications across the two systems, and off-diagonal cells represent divergent classifications.

Precision. Precision is an indicator of how accurately a system has identified instances of a specified sentiment category. Precision is defined as

$$P = \frac{TP}{TP + FP} \tag{3.11}$$

where TP is the number of True Positives and FP is the number of False Positives (Google for Developers, n.d.). These variables can be obtained via the confusion matrix of the two systems. Precision is a crucial metric for the comparative framework, as it gives insight into how well **System B** classifies a given sentiment category relative to **System A**.

Recall. Recall is an indication of whether a system can find all the references to a specific sentiment category. Recall is defined as

$$R = \frac{TP}{TP + FN} \tag{3.12}$$

where TP stands for the number of true positive classifications and FN is the number of false negative classifications. Recall is crucial to the comparative framework as it allows us to see how well **System B** classifies a negative sentiment category relative to **System A**.

F1-Score. The F1-Score merges Precision and Recall into one metric by taking their harmonic mean. This score gives a well-rounded view of how well a classification model is performing, factoring in both false positives and false negatives (scikit-learn developers, n.d.):

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \tag{3.13}$$

The F1-Score will be calculated for each sentiment category: Negative, Neutral, and Positive.

Macro-F1. Macro-F1 will be used to determine the overall classification performance across the three sentiment categories. It calculates the F1-Score for each category and gives equal weight to each category:

$$\text{Macro-}F_1 = \frac{1}{3} \sum_{c=1}^{3} F_{1,c} \tag{3.14}$$

A higher Macro-F1 score signals that both the precision and recall of the classification model are performing well across all three sentiment categories. Macro-F1 allows the comparative framework to give insight into the overall classification performance measure of **System B** against **System A**, covering all three ternary categories: Negative, Positive, and Neutral—highly relevant for an event like a natural disaster, which can evoke a broad

range of sentiment.

Cohen's Kappa. Cohen's Kappa will be used to evaluate the agreement between the classifications of the two systems. This metric takes into account the possibility of agreement by chance (McHugh, 2012):

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{3.15}$$

where p_o represents the observed proportion of agreement and p_e represents the expected proportion of agreement by chance. This provides a comparative framework to assess confidence that the insights from the performance measures are not merely due to chance for both **System A** and **System B**.

Acceptability Thresholds

The values for Reference-Relative Precision, Recall, F1-Score, Macro-F1, and Cohen's Kappa for both **System A** and **System B** will be evaluated with **System A** as the reference classification. These values will then be interpreted with acceptability thresholds to reach a conclusion on their classification performance.

A Reference-relative F1-Score and Macro-F1 of at least 0.70 will be considered an acceptable level of classification performance, indicating that **System B** has attained enough balance between precision and recall relative to **System A** across sentiment categories. This threshold is adequate for noisy domains with ambiguous ground truths; a threshold of 0.70 is deemed strong in this context (Openlayer, 2026).

A Cohen's Kappa of at least 0.60 will be considered acceptable inter-system agreement between **System A** and **System B**, as it is considered the threshold for adequate agreement, and confidence may be placed in the outcomes (McHugh, 2012). These performance measures will be evaluated for interpretation of whether **System B** can produce classification outcomes in alignment with **System A**.

3.6.2 Research Question 2: Joint Matrix Latent Topic Structure and Sentiment Classification Relationship Analysis

Regarding Research Question 2, an assessment is conducted to determine whether a relationship exists between the respective joint matrices of **System A** and **System B** and how strong the association is between their joint matrices' latent topic structures and sentiment classifications. This must be done to analytically justify that the two systems have the ability to capture the latent topics and sentiments of Facebook posts during natural disasters, and their quantifiable measures of performance in doing so.

Bivariate Analysis. As a statistical method, bivariate analysis serves as an analytical tool that determines the empirical relationship between two variables, their interaction with one another, trends, and possible causal relationships. It aims at examining the relationship between two variables to gain insight into whether such a relationship exists, how strong it is, and whether it is positive or negative. By examining this, it allows a quantifiable understanding of two variables (Ben, n.d.). Since our joint matrix is composed of latent topics and sentiment classifications, this can be interpreted as two variables that the study seeks to analytically determine whether their joint matrix is significantly associated.

Within this investigation, the system-generated latent topics and sentiment classifications from the joint matrices of both systems will serve as the two variables evaluated for bivariate analysis. Prior to performing further bivariate analysis using the Chi-Square Statistic and Cramer's V, the output must be in the form of a cross-tabulation to create a structured framework of computation in matrix format. The joint matrix itself is a form of bivariate analysis, as it is a cross-tabulation or contingency table that allows us to see the frequency distribution of both variables in the same format and gives insight into their associations (Ben, n.d.).

Chi-Square Statistic Test. The chi-square test is used to determine whether there is a significant difference between observed frequencies and expected frequencies and, in contingency tables, the relationship between categorical variables. It can be used to determine whether a significant relationship exists between two or more variables. This association can be negative, positive, or an absence of association. To test whether an association exists, we set the null hypothesis that no association exists and the alternative hypothesis that an association exists. To decide whether to reject or fail to reject the null hypothesis, we first compute the chi-square statistic (χ^2) and obtain the critical value to determine whether we conclude that no association exists or that an association exists (Singh, 2020). In this study, we state that:

- H_0: There is no relationship between what crisis issue is discussed (Topic) and the emotion behind it (Sentiment).
- H_1: There is a systematic relationship between topics and sentiments, which may indicate that the semantic profiles of Facebook posts align with their sentiment profiles.

The formula to attain χ^2 is

$$\chi^2 = \sum_{i=1}^{R} \sum_{j=1}^{C} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \tag{3.16}$$

where R is the number of rows of the joint matrix (the latent topics), C is the number of columns of the joint matrix (the sentiment classifications), O_{ij} are the observed frequencies in the joint matrix—the actual raw count of Facebook posts that either of the two systems assigns to latent topic i and sentiment classification j—and E_{ij} are the expected frequencies, or the calculated marginal totals of the matrix if the null hypothesis is true. To attain E_{ij}, the formula is

$$E_{ij} = \frac{R_i\, C_j}{n} \tag{3.17}$$

For determining thresholds, we need to test χ^2 against the chi-square distribution table;

to attain that, we first need the degrees of freedom df:

$$df = (R - 1)(C - 1) \tag{3.18}$$

After this, a Python function is used to determine the critical value; it takes the contingency table as its input, and if the associated p-value is less than the chosen significance level, the null hypothesis is rejected and the joint matrix is deemed to have a significant association (Singh, 2020).

Cramer's V. Cramer's V is utilised to test the strength of association on contingency tables of two variables that are more than binary in possible frequencies. It is an expansion of Phi (ϕ) for two variables that exceed binary frequencies. It inherits the same interpretability of results, which is 0 for no association to 1 for complete dependence. The formula is

$$V = \sqrt{\frac{\chi^2/n}{\min(R - 1,\ C - 1)}} \tag{3.19}$$

which yields the V used to determine whether the joint matrix of the two systems possesses a weak or strong association (Ben-Shachar et al., 2023). To determine whether the relationship between the two variables has strong dependence, we attain the degrees of freedom of the smaller dimension of the joint matrix:

$$df^* = \min(R - 1,\ C - 1) \tag{3.20}$$

As the sentiment classification will be the smallest dimension—assuming the latent topics exceed 3—$df^* = 2$, and the thresholds to interpret the results, given a V, are shown in Table 3.3.

This gives us an acceptable threshold that will deem the joint matrices of the two systems as having a significant strength of association: V must exceed 0.07 (CASRAI, 2026).

Table 3.3: Interpretation Thresholds for Cramer's V

df^*	**Small**	**Medium**	**Large**
2	0.07	0.21	0.35

3.6.3 Research Question 3: Blind Stakeholder Evaluation

Research Question 3 evaluates the stakeholder assessment regarding the outputs generated by **System A** and **System B**. From the common evaluation corpus, a randomly selected sample will be examined through a blind evaluation procedure in which the order of the systems' outcomes will be randomized. A simple majority rule will determine the aggregation rule for each document, because the identity of the system making each output will not be revealed to the evaluators. When this evaluation takes place, agreement, differences in ratings, and paired assessment outcomes will be considered. Each evaluator is expected to evaluate 50–100 randomly sampled documents drawn from the outputs of **System A** and **System B**.

Agreement Rate. Agreement Rate will be utilized to determine the proportion of stakeholder evaluations in which the assessment agrees with the specified reference or comparison outcome. Agreement Rate is calculated as

$$AR = \frac{N_{\text{agree}}}{N_{\text{total}}} \times 100\% \tag{3.21}$$

A higher Agreement Rate indicates greater consistency among the stakeholder assessments.

McNemar's Test. For stakeholder evaluation, paired binary results will be examined using McNemar's Test, which is suitable for systematic comparisons when identical assessors evaluate two systems in two categories. The test statistic incorporates continuity correction:

$$\chi^2_{\text{McNemar}} = \frac{(|b - c| - 1)^2}{b + c} \tag{3.22}$$

In the calculation, b indicates cases where **System A** succeeds while **System B** fails, and c indicates cases where **System B** succeeds while **System A** fails. This comparison aims to reveal operational differences, with the p-value being analyzed to assess the statistical significance of the differences between the outcomes of the two systems.

Acceptability Thresholds

The resulting performance measures will be examined to confirm whether a statistically significant difference exists between the paired outcomes of **System A** and **System B** when assessing overall performance.

An Agreement Rate of at least 70% is deemed acceptable as high agreement in inter-rater reliability (Stemler, 2004). Meanwhile, a McNemar's Test at the 0.05 significance level will be the minimum acceptability threshold (MCP Analytics, 2025); that is, a system is representative of the sentiments of stakeholders around natural disasters in Bulacan and the CAMANAVA area, as it indicates no significant differences between the system and the sentiment of stakeholders around the documents. These performance measures will be applied independently to both **System A** and **System B** before a relative comparative analysis that evaluates their relative performance with one another.

3.7 Respondents and Expert Evaluators of the Study

The development and comparative evaluation of the SAKLOLO. natural language processing pipeline requires highly specialized domain expertise rather than general population sampling. Consequently, the study utilizes a purposive sampling design to form a Blind Evaluation Stakeholder Panel.

3.7.1 Exclusion of Population-Based Sample Size Formulas

When determining the number of evaluators for natural language processing (NLP) architectures, traditional probabilistic calculations (such as Cochran's or Slovin's formulas) are

mathematically and methodologically incompatible with the research design. These formulas are designed for epidemiological or sociological surveys where the goal is to achieve a representative demographic cross-section of a generalized population. They assume that every member of the population has an equal probability and capacity to provide valid data.

The evaluation of the proposed framework, however, is not a demographic survey but rather an experimental computational benchmark. The study requires human evaluators to audit the classification outputs of a hybrid pipeline (LDA + XLM-RoBERTa) and an open-weight transformer (`Qwen3-1.7B-Instruct-Q4_K_M`).

Evaluating contextual embeddings, coherence, and algorithmic classification disparities requires calibrated judgment from domain specialists. Using a purposively selected expert panel provides the controlled conditions necessary to execute specific non-parametric statistical validations (such as McNemar's test and rank correlation coefficients) to detect genuine differences in model configurations without introducing data dilution.

3.7.2 Blind Evaluation Stakeholder Panel (System Evaluation)

To execute the blind comparative evaluation between the lightweight hybrid model (**System A**: LDA + XLM-RoBERTa) and the state-of-the-art open-weight (**System B**: `Qwen3-1.7B-Instruct-Q4_K`
a total panel of 10 to 20 evaluators will be recruited. This panel is divided equally into two specialized cohorts.

In software engineering and machine learning benchmark literature, evaluating models against human baselines relies on expert panel auditing rather than large-scale probability sampling. A focused panel of calibrated domain specialists provides the necessary statistical power to execute paired nonparametric tests—specifically McNemar's test for classification disparity, preference ratio calculations, and Cohen's Kappa for inter-system consensus—while eliminating evaluation fatigue and ensuring reliable judgments.

Table 3.4: Cohort A and Cohort B Descriptions

Cohort A: DRRM Operational Officers (n = 5–10)	**Cohort B: Academic and Technical Experts** (n = 5–10)
Source. Active personnel, incident commanders, and communication dispatchers from the Provincial and Municipal Disaster Risk Reduction and Management Offices (DRRMOs) of the Bulacan and CAMANAVA corridors.	*Source.* Faculty members and researchers specializing in Applied Mathematics, Computer Science, and Data Analytics, specifically targeting the Mathematics Department of the College of Science and the allied faculty of the CICT at Bulacan State University.
Operational Role. These practitioners evaluate the practical applicability, operational fidelity, and functional alignment derived from the models without knowing which system generated the output. They provide judgment on whether the extracted sentiment and disaster urgency accurately reflect actual field conditions.	*Operational Role.* These experts assess the mathematical validity and thematic coherence of the extracted topic probabilities and contextual embeddings, ensuring the algorithms operate within accepted standards of information theory and machine learning rigor.

Table 3.5: Summary of Study Participants

Participant Group	**Target Size (n)**	**Sampling Method**	**Target Affiliation**	**Research Function**
DRRM Practitioner Panel	5–10	Purposive	Bulacan and CAMANAVA DRRM Offices	Blind evaluation of system outputs; assesses operational applicability and severity relevance.
Academic Technical Panel	5–10	Purposive	Mathematics and Computer Science Faculty (College of Science / CICT)	Technical validation of thematic coherence, joint embeddings, and comparative algorithmic quality.
Total Panel	**10–20**	**Purposive**	—	—

3.8 Research Instrument

The research instrument of the study is a structured blind evaluation form. It is meant to assess the agreement of stakeholder assessments with the outputs of **System A** and **System B**. The instrument is designed to allow both systems to be evaluated independently using the same sampled disaster-related textual documents while concealing the identity of the producing system.

The stakeholder will assess whether the two presented sentiment classifications agree with their own assessment of the document using a check-or-cross response format. The same procedure is to be applied to the output of the other system in the same format. The evaluation form records the stakeholder's assessment for each system output and the corresponding identifiers. These responses are to be converted into agreement observations for statistical analysis. The instrument is to be administered with the same instructions, response categories, and evaluation conditions for all participating stakeholders to maintain consistency across observations.

3.9 Ethical Considerations

This study observes ethical considerations for the safety of participants and maintains the integrity of this paper throughout the development of *SAKLOLO.: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA*. The researchers will follow ethical guidelines such as collecting data from identified sources, obtaining consent, ensuring the confidentiality of participants and voluntary participation, using social media responsibly, and maintaining integrity in presenting the research results for transparency. Appropriate procedures are established to maintain confidentiality and to protect the privacy of all participants in this study. The study adheres to the following ethical principles:

Data Privacy and Confidentiality.

Republic Act No. 10173, or the Data Privacy Act of 2012, establishes the rights of individuals needed to protect their personal information, and the organization is responsible for the personal information of participants (Republic of the Philippines, 2012). The needed disaster-related report data will be processed only if needed, solely for this study. The study will exclude unnecessary personal information. The study will focus on collecting disaster-related information from identified sources such as social media and DRRM reports in Bulacan and CAMANAVA. This data will be filtered to remove irrelevant data and personal information for the privacy and accuracy of the study. All collected data are for academic research and model validation purposes only (National Privacy Commission, 2016).

Voluntary Participation and Informed Consent.

Participants will be informed about the purpose and procedure of this study prior to their involvement. Participants in this study are voluntary; they have the right to decline to participate in this study without penalty. To ensure accuracy and prevent biases during evaluation, a protocol will be established wherein evaluators will be anonymized without being informed whether the predictions were from the LLM baseline model or the hybrid pipeline system (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

Confidentiality of Participants.

The identity of participants and personal information of stakeholders will be kept confidential in this study. The evaluation process will use a blind evaluation procedure so that the system does not reveal the identity of evaluators. Possible data that can identify participants will be excluded from the results to prevent biases and to protect the participants. In accordance with the Data Privacy Act of 2012 (National Privacy Commission, 2016; Republic of the Philippines, 2012), the identity of participants shall be protected against alteration and unauthorized access.

Responsible Use of Social Media Data.

Only data relevant to the study will be collected, such as disaster-related data from publicly accessible social media of DRRM offices. The usernames of commenters from social media and other personal identifiers will be excluded from the data collection process. The researchers ensure that social media will be used responsibly and solely for academic purposes, not to harm anyone. Moreover, all collected data from social media will be anonymized and processed in accordance with ethical research standards (National Privacy Commission, 2016).

Research Integrity.

The researchers of this study uphold transparency, honesty, and objectivity throughout the development of the study. All data related to this study, such as evaluations and results, will be properly collected and studied accurately without falsification and fabrication. This study will be developed ethically and honestly, and results will be presented with integrity whether the outcome is expected or not (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979). This study ensures reliability, credibility, and transparency. Moreover, citations to all sources such as DRRM reports, social media posts, and other materials will be properly acknowledged within the paper.

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